

RABIN KHADKA

Data Science | Business Analytics | Explainable AI | Supply Chain & Decision-Making

Winston-Salem, NC | +1 (646) 300-5525 | [Email](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

RESEARCH PROFILE

Interdisciplinary researcher with academic training in business administration and data science, professional analytics experience, and prior classroom teaching. Research interests include machine learning, explainable AI, forecasting, human and managerial use of analytical models, business analytics, and supply-chain decision-making. Interested in research that connects quantitative methods with practical decision problems and supports a long-term career in university teaching and research.

RESEARCH INTERESTS

Machine Learning • Explainable AI • Forecasting & Time Series • Human-AI Decision Support • Business Analytics • Supply Chain & Operations Analytics • Managerial Decision-Making

EDUCATION

Carolina University - Winston-Salem, North Carolina

May 2024

Master of Science in Data Science | GPA: 3.83/4.00 | Magna Cum Laude

Selected Coursework: Statistical Modeling; Big Data Analytics; Machine Learning Algorithms; Advanced Database Management Systems

Texas College of Management and IT - Kathmandu, Nepal

Mar 2020

Bachelor (Hons) in Business Administration | GPA: 3.90/4.00 | Degree awarded by Lincoln University College, Malaysia

Selected Coursework: Quantitative Methods; Management Information Systems; Business Research Methods; Logistics and Supply Chain Management; Marketing Research

RESEARCH MANUSCRIPT

Khadka, R. (2026). Explainable Artificial Intelligence in Demand Forecasting: From Model Explanation to Managerial Decision Support. *Independent focused review manuscript.*

SELECTED RESEARCH & ANALYTICAL PROJECTS

Upstream Price Pressures and Consumer Inflation: U.S. Supply Chain Price Transmission

Python | Time Series

- Analyzed monthly U.S. CPI, Producer Price Index, and Import Price Index data from 2015-2026 to examine upstream-to-downstream price relationships using time-series analysis, cross-correlation, and Granger-causality testing.
- Used official U.S. government series and documented the analysis as an independent research project.

Amazon Product Analytics: Sentiment Analysis & Predictive Modeling

Python | NLP | ML

- Analyzed 1,465 Amazon product records covering prices, discounts, ratings, product categories, and customer-review text through data cleaning, exploratory analysis, sentiment analysis, and predictive modeling.
- Applied TextBlob and VADER sentiment measures and compared linear and ensemble regressors for discount prediction; tuned Gradient Boosting produced the strongest test R^2 (0.3223).

[GitHub repository](#)

SELECTED RESEARCH & ANALYTICAL PROJECTS (CONTINUED)

Global Life Expectancy Prediction

Python | ML

- Evaluated demographic, economic, and health indicators as predictors of life expectancy using multiple regression, decision trees, and random forests; compared models using MAE, MSE, and R^2 .
- Random Forest produced the strongest predictive performance ($R^2 \approx 0.96$), substantially outperforming the tested regression and single-tree approaches.

Insurance Fraud Detection Using Machine Learning

Python |
Classification

- Prepared an imbalanced insurance-claims dataset using feature engineering, scaling, and SMOTE; compared Random Forest, Gradient Boosting, SVM, and neural-network approaches.
- Evaluated model generalization using train/test metrics and cross-validation, with attention to class imbalance and overfitting.

PROFESSIONAL & TEACHING EXPERIENCE

Junior Social Media Data Analyst | Kaebox LLC - Winston-Salem, NC

Apr 2025-Present

- Build Python- and SQL-based workflows to ingest, clean, structure, and analyze digital engagement and text data for KPI monitoring and trend analysis.
- Apply regression, clustering, and classification methods to market and engagement data for consumer-behavior assessment and brand-sentiment analysis.
- Develop Tableau dashboards that translate performance indicators into decision-oriented business insights for nontechnical stakeholders.

Business Organization Instructor | Wits Academy - Boudha, Kathmandu, Nepal

Apr 2020-Mar 2022

- Taught Business Organization to Grade 11 and Grade 12 students, including management, organizational structure, business operations, and decision-making.
- Prepared lessons, assessments, and instructional materials and guided students in applying business concepts to practical situations.

TECHNICAL & QUANTITATIVE SKILLS

Programming & Data: Python, R, SQL, pandas, NumPy, scikit-learn, statsmodels

Machine Learning & Statistics: Regression, classification, random forests, gradient boosting, SVM, Naive Bayes, PCA, cross-validation, hyperparameter tuning, time-series analysis

Deep Learning & Text Analytics: TensorFlow/Keras, multilayer perceptrons, NLP, sentiment analysis, TextBlob, NLTK/VADER

Visualization & Data Systems: Tableau, Matplotlib, Seaborn, database management, API data extraction

ACADEMIC & CAREER OBJECTIVE

Pursue doctoral research that combines quantitative modeling with questions about how predictions and analytical information are understood and used in real decisions. Areas of particular interest include explainable AI, forecasting, business and supply-chain analytics, and human or managerial decision support. Long-term goal is to build an academic career as a university professor and researcher.